Course Number: CGN 5432
Submitter: Mostafa Sameer,
P.ID: 6513997
Ph.D. Student, Department of Civil Engineering

Instructor: Dr. ***Omar Nofal****,*
Assistant Professor, Department of Civil and Environmental Engineering.

The main purpose of this research is to evaluate flood risk and chance of exceedance based on different flood depths. The study creates cumulative distribution functions (CDF) using DS0, DS1, DS2, DS3, and DS4 datasets to understand flood patterns in a region. This helps predict flood depths over time, which can inform flood management, infrastructure development, and emergency preparedness.

Nueces County is situated in southeastern Texas, USA, with Corpus Christi as its county seat. Coastal flooding is common in Corpus Christi and other parts of the county due to its closeness to the Gulf of Mexico. Flood risk analysis is essential to Nueces County's disaster management and urban development due to its vulnerability to hurricanes and other extreme weather.

Public policy, risk reduction, and infrastructure development depend on flood depth analysis in this region. Understanding flood risk can help with zoning, development, and floodplain management for a big coastal population. This study could also assist local authorities prioritize flood responses to reduce property damage and protect residents.

Introduction: The main purpose of this research is to evaluate flood risk and chance of exceedance based on different flood depths. The study creates cumulative distribution functions (CDF) using DS0, DS1, DS2, DS3, and DS4 datasets to understand flood patterns in a region. This helps predict flood depths over time, which can inform flood management, infrastructure development, and emergency preparedness. This report outlines the flood hazard assessment and building risk evaluation conducted for Nueces County, Texas. The objective is to identify flood-prone areas, evaluate the potential damage to structures, and determine the vulnerability of buildings to flood events. The data used includes building locations, elevations, flood depths, and damage state probabilities.

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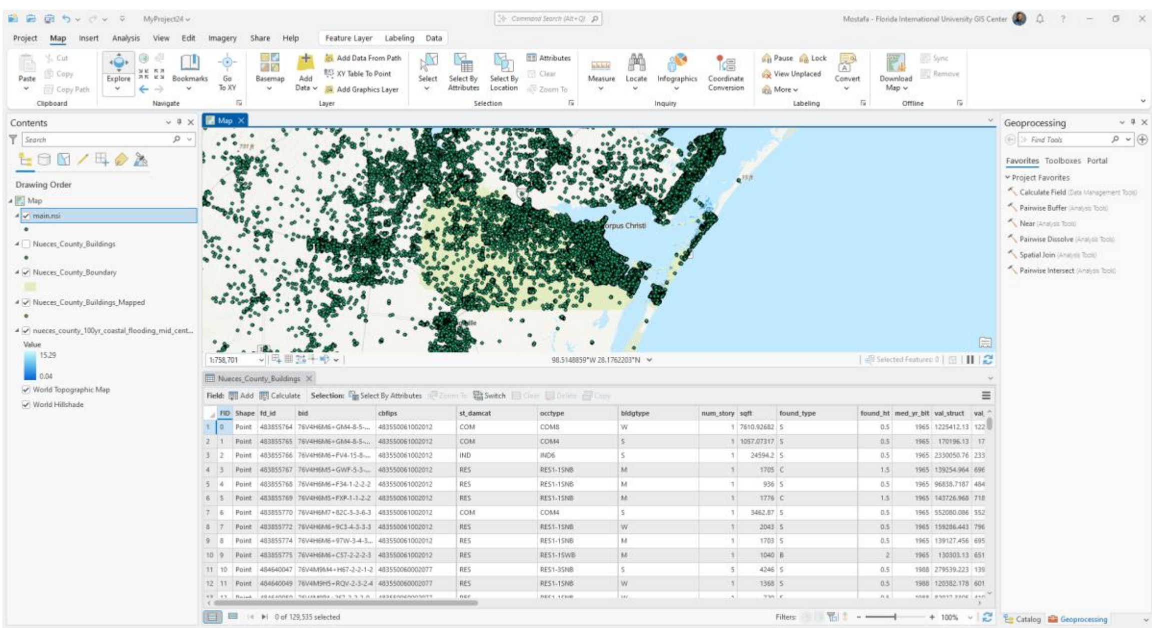
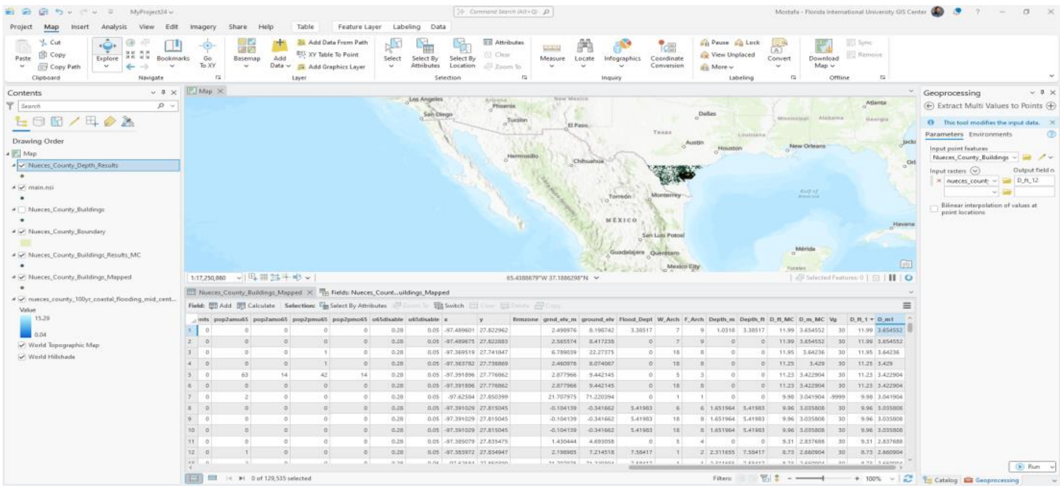
1. GIS Map of Nueces County

The first map represents the building distribution within Nueces County. The buildings are plotted as points, overlaid on the map to highlight areas at risk of flooding. The data is derived from a shapefile containing detailed building attributes.



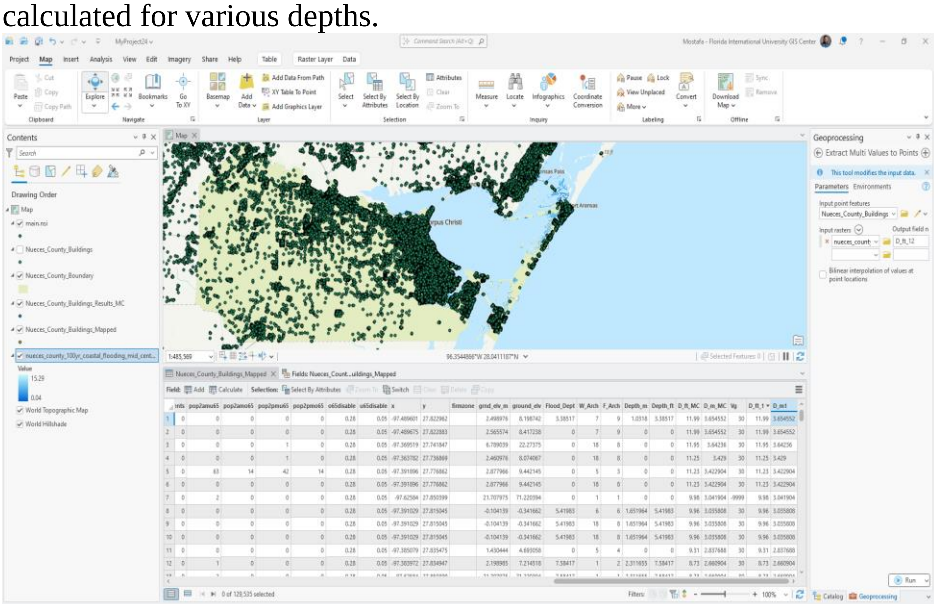
2. Flood Hazard Map:

The map shows the coastal flooding risk within Nueces County, emphasizing areas affected by storm surge and tidal flooding.



3. Depth Evaluation:

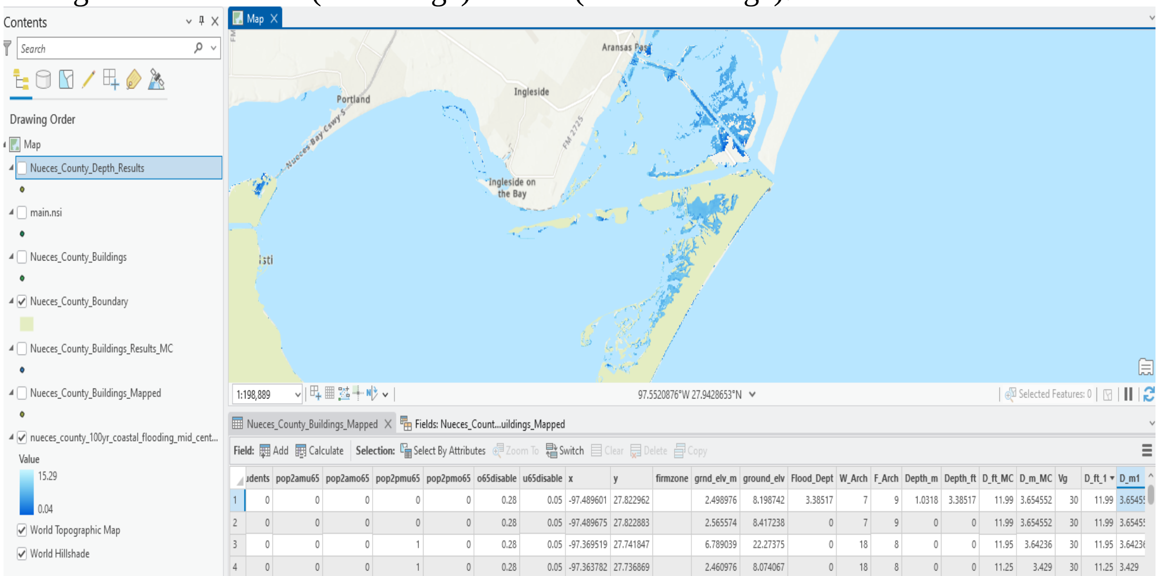
In the figure, the flood depth results are shown for each building. These are essential for understanding the severity of flooding in each area, with the water surface elevation (WSE) calculated for various depths.



D_ft_1	D_m1
11.99	3.654552
11.99	3.654552
11.95	3.64236
11.25	3.429
11.23	3.422904
11.23	3.422904
9.98	3.041904
9.96	3.035808
9.96	3.035808
9.96	3.035808
9.31	2.837688
8.73	2.660904

4. Risk Evaluation of Buildings:

The figure below demonstrates the building damage states based on flood depths and corresponding probabilities. The buildings are color-coded according to their assessed damage state from zero (no damage) to four (severe damage).



5. Risk by Damage State:

This table shows the calculated probabilities for each building's damage state, considering the building's elevation and flood depth. The values indicate the likelihood of a structure sustaining varying levels of damage.

Nueces_County_Buildings_Mapped | Fields: Nueces_Count...uildings_Mapped

	udents	pop2amu65	pop2amo65	pop2pmu65	pop2pmo65	o65disable	u65disable	x	y	firmzone	grnd_elv_m	ground_elv	Flood_Dept	W_Arch	F_Arch	Depth_m	Depth_ft	D_ft_MC	D_m_MC	Vg	D_ft_1	D_m1
1	0	0	0	0	0	0.28	0.05	-97.489601	27.822962		2.498976	8.198742	3.38517	7	9	1.0318	3.38517	11.99	3.654552	30	11.99	3.6545
2	0	0	0	0	0	0.28	0.05	-97.489675	27.822883		2.565574	8.417238	0	7	9	0	0	11.99	3.654552	30	11.99	3.6545
3	0	0	0	1	0	0.28	0.05	-97.369519	27.741847		6.789039	22.27375	0	18	8	0	0	11.95	3.64236	30	11.95	3.6423
4	0	0	0	1	0	0.28	0.05	-97.363782	27.736869		2.460976	8.074067	0	18	8	0	0	11.25	3.429	30	11.25	3.429
5	0	63	14	42	14	0.28	0.05	-97.391896	27.776862		2.877966	9.442145	0	5	3	0	0	11.23	3.422904	30	11.23	3.4229
6	0	0	0	0	0	0.28	0.05	-97.391896	27.776862		2.877966	9.442145	0	18	8	0	0	11.23	3.422904	30	11.23	3.4229
7	0	2	0	0	0	0.28	0.05	-97.62584	27.850399		21.707975	71.220394	0	1	1	0	0	9.98	3.041904	-9999	9.98	3.0419
8	0	0	0	0	0	0.28	0.05	-97.391029	27.815045		-0.104139	-0.341662	5.41983	6	6	1.651964	5.41983	9.96	3.035808	30	9.96	3.0358
9	0	0	0	0	0	0.28	0.05	-97.391029	27.815045		-0.104139	-0.341662	5.41983	18	8	1.651964	5.41983	9.96	3.035808	30	9.96	3.0358
10	0	0	0	0	0	0.28	0.05	-97.391029	27.815045		-0.104139	-0.341662	5.41983	18	8	1.651964	5.41983	9.96	3.035808	30	9.96	3.0358
11	0	0	0	0	0	0.28	0.05	-97.385079	27.835475		1.430444	4.693058	0	5	4	0	0	9.31	2.837688	30	9.31	2.8376
12	0	1	0	0	0	0.28	0.05	-97.383972	27.834947		2.198985	7.214518	7.58417	1	2	2.311655	7.58417	8.73	2.660904	30	8.73	2.6609

Flood risk analysis based on different datasets helps to understand the varying probabilities of exceedance under different flood conditions. Each archetype provides a unique perspective on flood risk, from severe urban flooding (DS0, DS1, DS2, DS3, DS4) to areas with manageable risks but still susceptible to significant flood events (DS0, DS1, DS2, DS3, DS4). Urbanization, infrastructure quality, geography, and climate play key roles in determining the flood risks in each region.

The distribution of these datasets highlights the importance of considering both local and regional flood management strategies. In areas like DS1, where flood risks are high even at low flood depths, there is an urgent need for substantial infrastructure investments, whereas in areas like DS4, focusing on the environmental factors influencing flood depth could prove beneficial.

These graphs show 15 Archetypes of the relationship between the probability of exceedance and flood depth for different datasets (DS1, DS2, DS3, and DS4). The curve indicates how the probability changes as the flood depth increases.

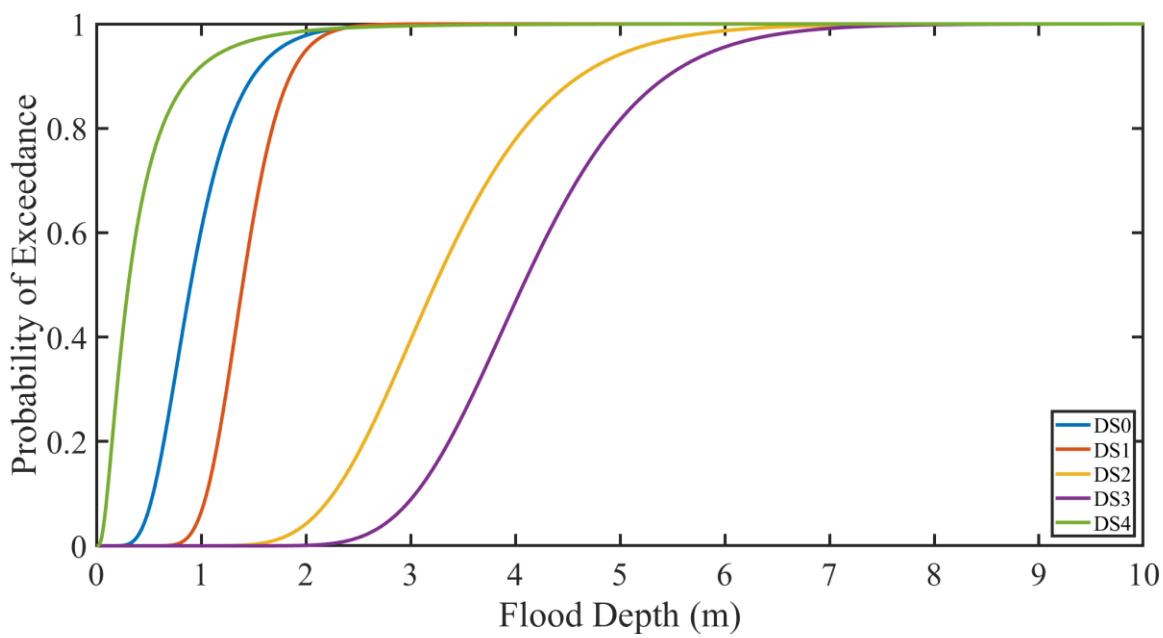


Figure 1: Probability of Exceedance vs Flood Depth for Archetype 1.

This graph shows a similar relationship as Figure 1 but with slight variations in the datasets.

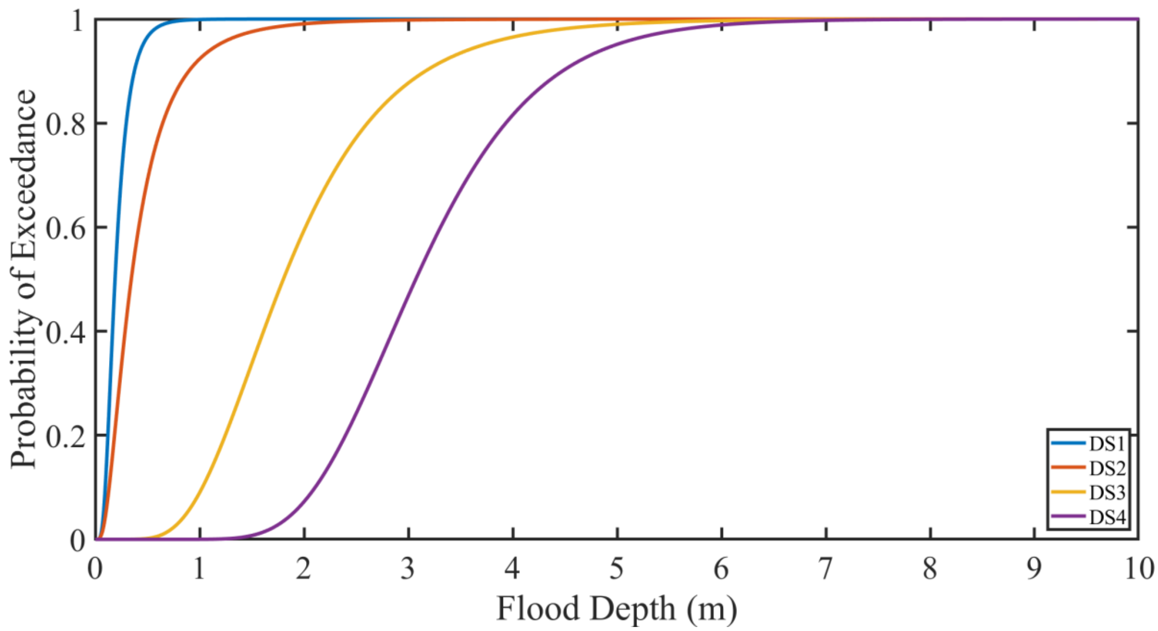


Figure 2: Probability of Exceedance vs Flood Depth for Archetype 2.

Another variation of the probability of exceedance versus flood depth, showing the effect of different datasets on the curve.

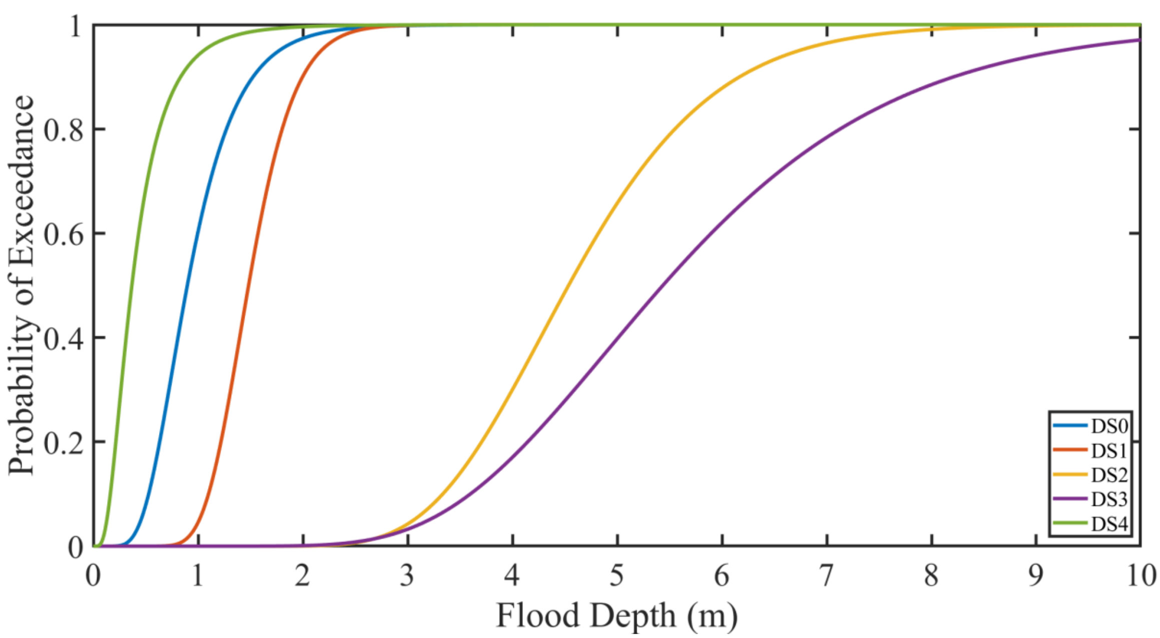


Figure 3: Probability of Exceedance vs Flood Depth for Archetype 3.

This plot illustrates the continuing trend of flood depth and probability of exceedance with yet another variation in the dataset.

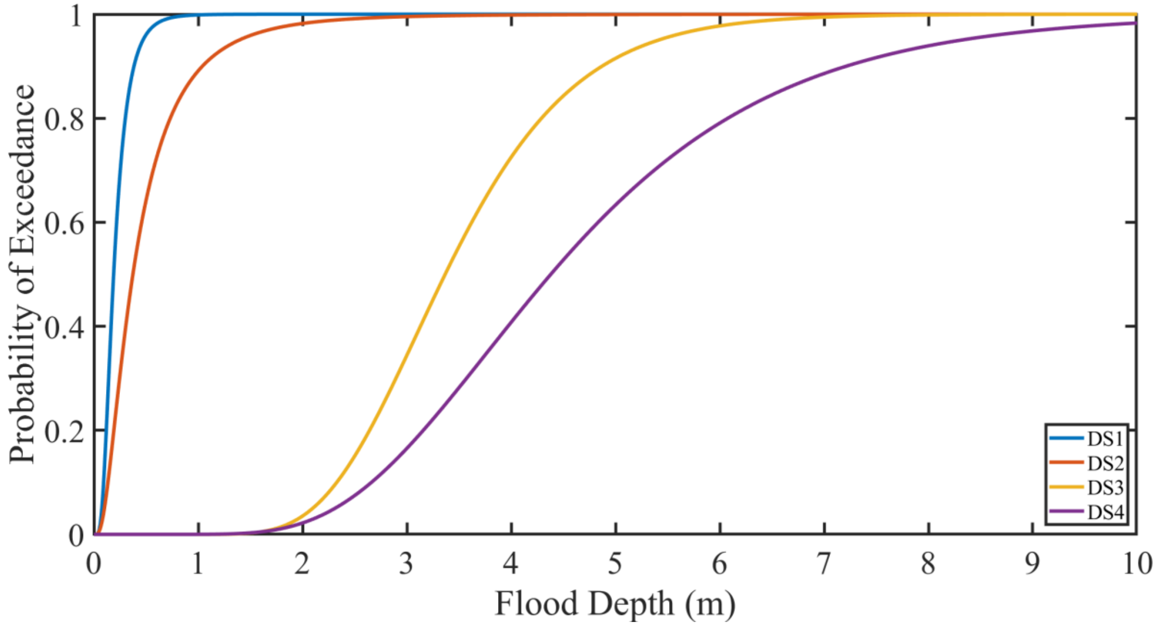


Figure 4: Probability of Exceedance vs Flood Depth for Archetype 4.

This figure adds more data points showing how the exceedance probability increases with depth.

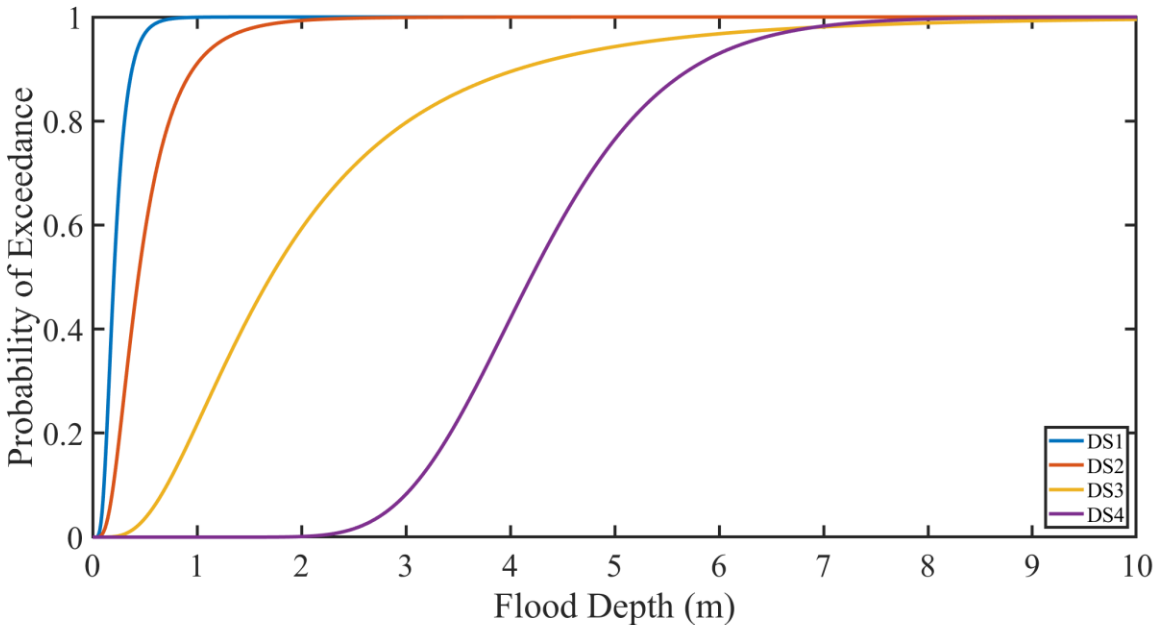


Figure 5: Probability of Exceedance vs Flood Depth for Archetype 5.

This figure showcases the shifting of curves with changes in the dataset.

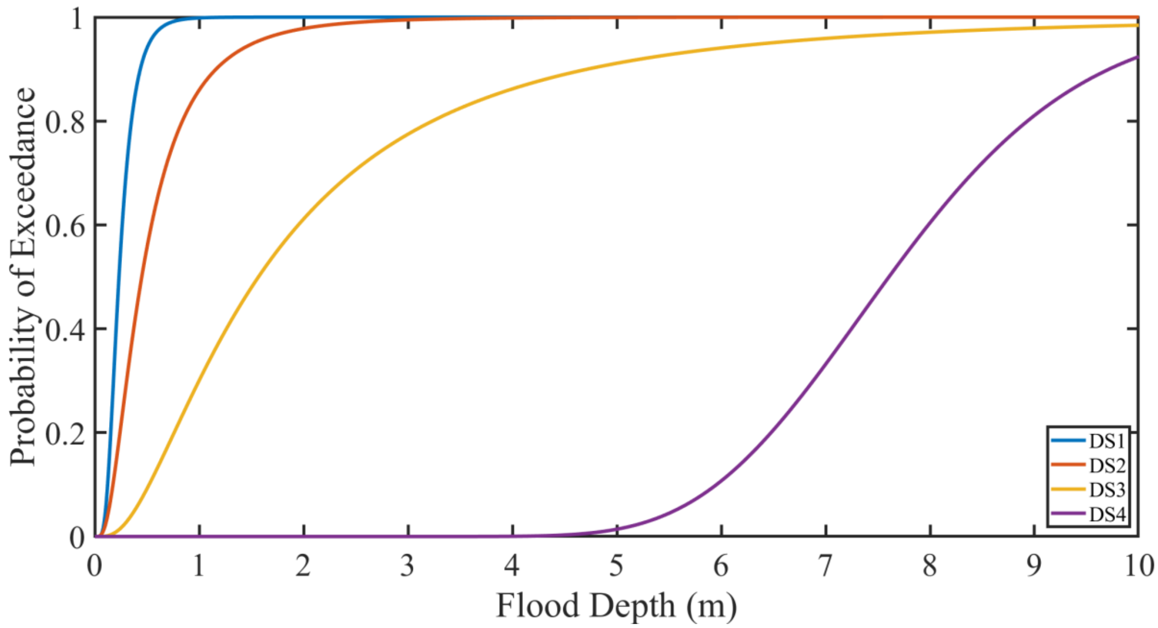


Figure 6: Probability of Exceedance vs Flood Depth for Archetype 6.

An additional figure showing how the exceedance probability behaves across different flood depths.

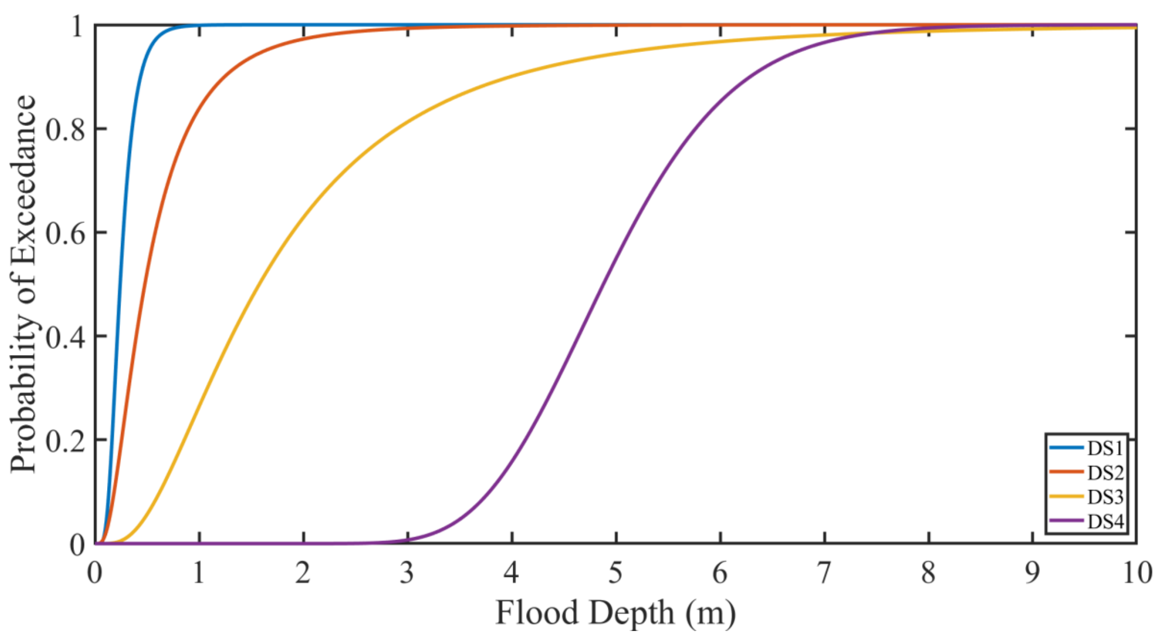


Figure 7: Probability of Exceedance vs Flood Depth for Archetype 7.

Further representation of the flood depth's effect on the probability of exceedance.

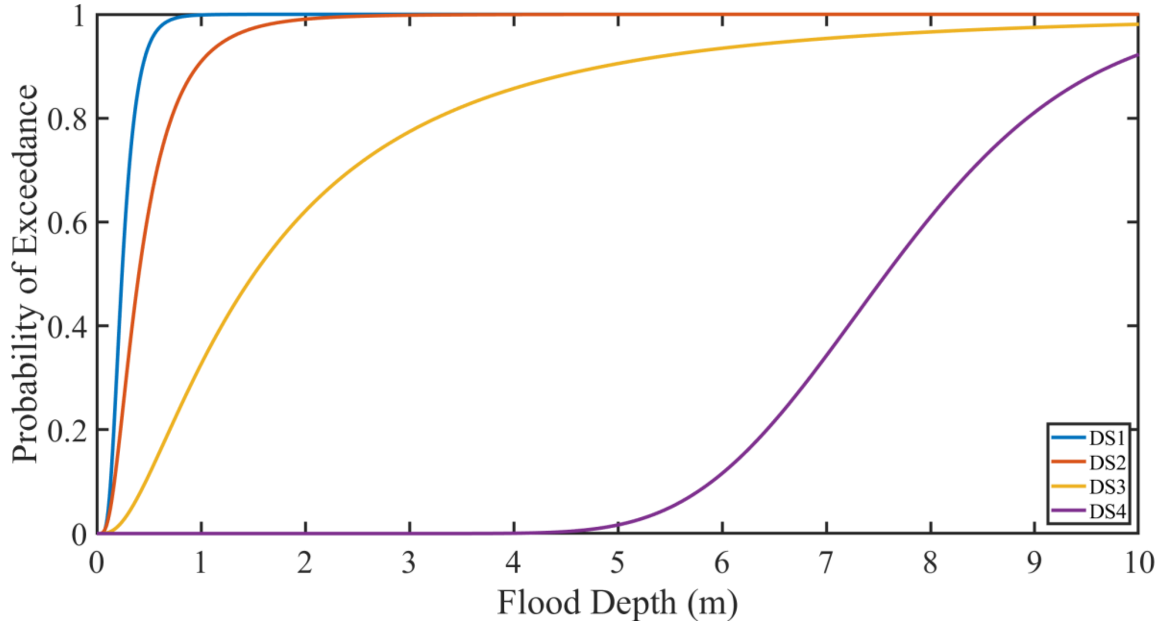


Figure 8: Probability of Exceedance vs Flood Depth for Archetype 8.

This graph highlights the difference in how the datasets impact the probability of exceedance.

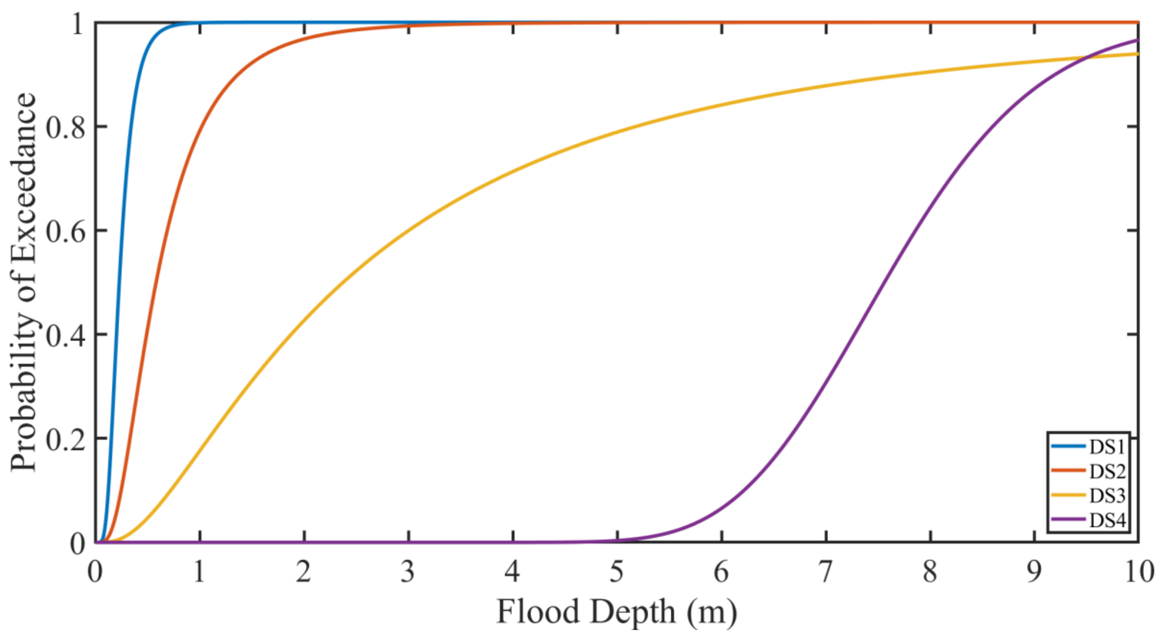


Figure 9: Probability of Exceedance vs Flood Depth for Archetype 9.

This figure shows the final comparison with all four datasets, indicating their overall trends in relation to flood depth.

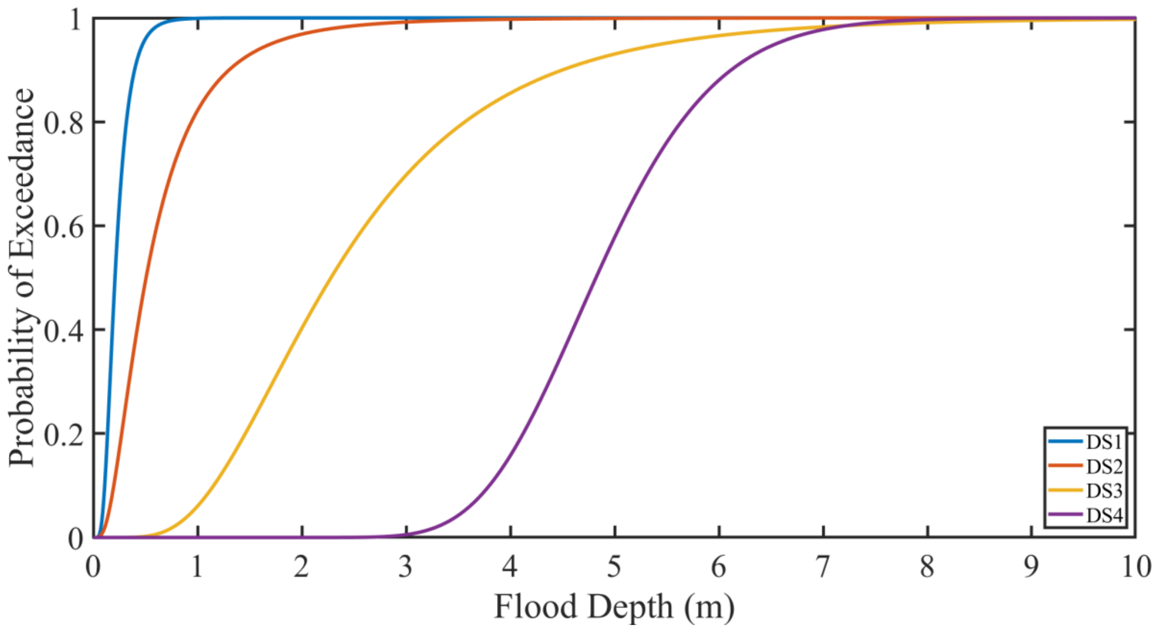


Figure 10: Probability of Exceedance vs Flood Depth for Archetype 10.

The figure demonstrates the accumulation of data from different sources, showing how various scenarios affect the flood depth versus exceedance probability curve.

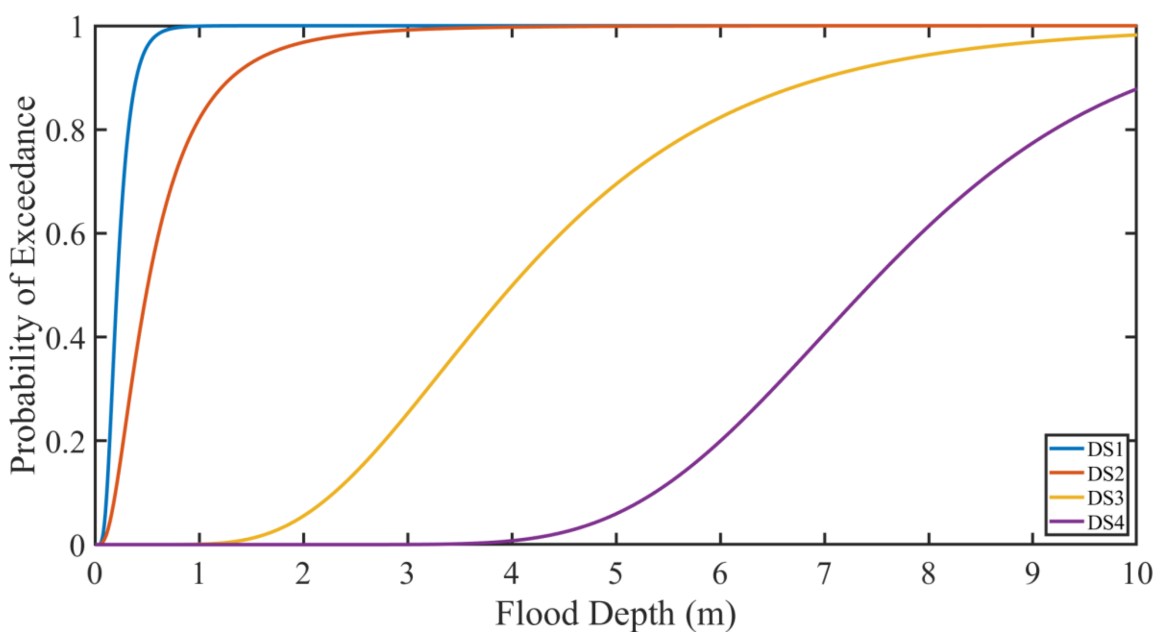


Figure 11: Probability of Exceedance vs Flood Depth for Archetype 11.

This graph further explores the relationship between flood depth and the likelihood of exceedance, offering a comparative analysis for all datasets.

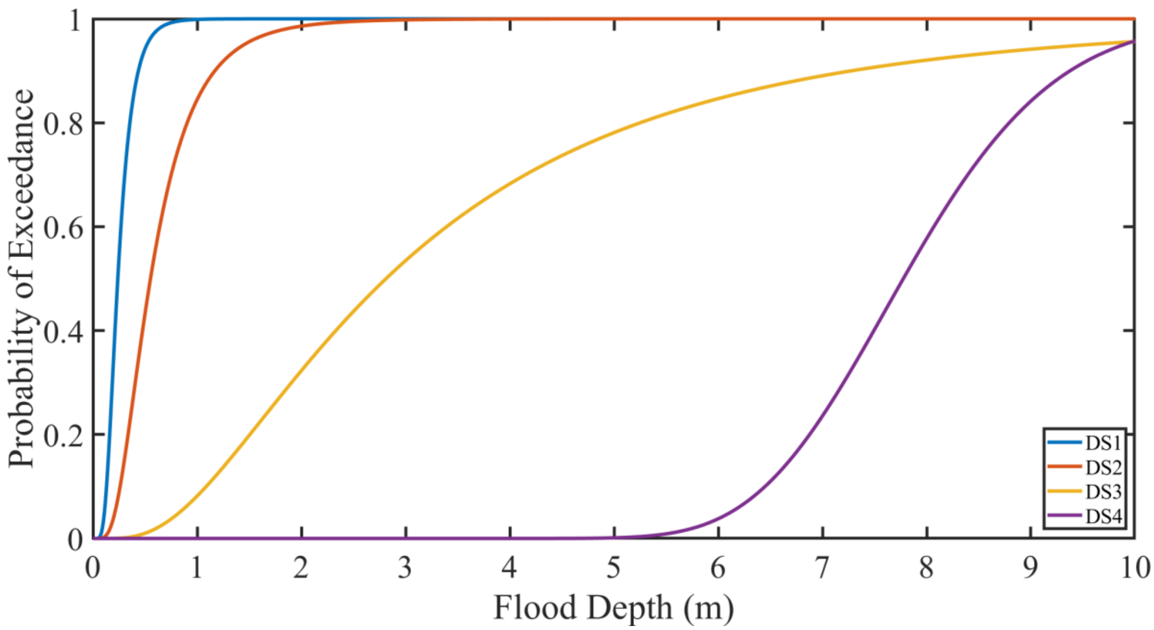


Figure 12: Probability of Exceedance vs Flood Depth for Archetype 12.

This final figure continues the comparative analysis between different datasets, highlighting the influence of various factors on the exceedance probability.

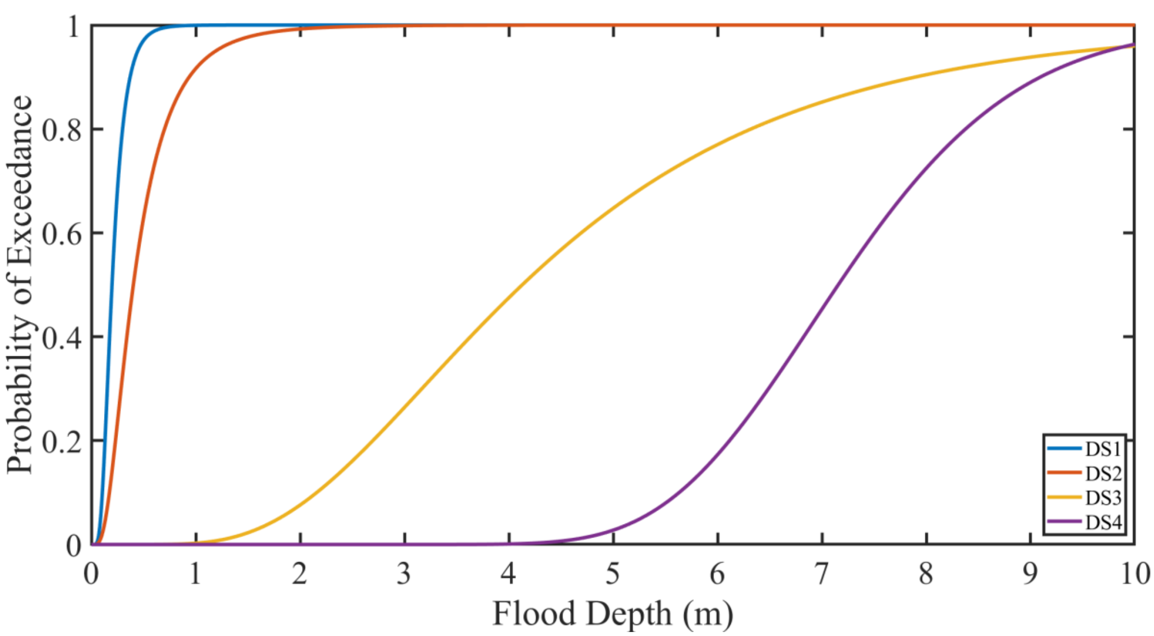


Figure 13: Probability of Exceedance vs Flood Depth for Archetype 13.

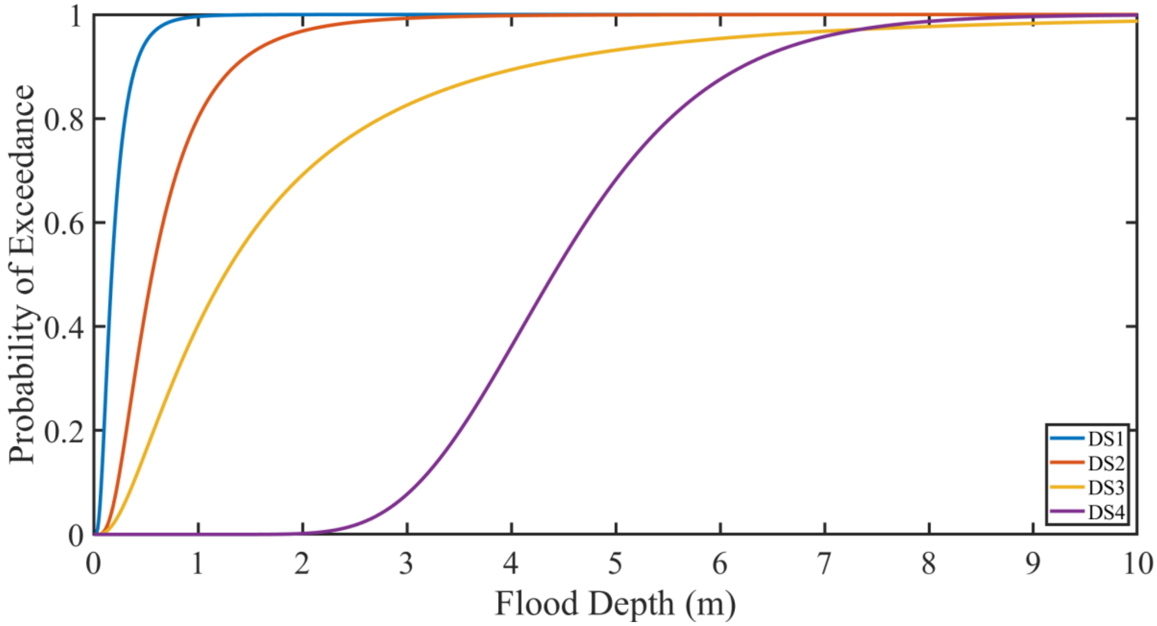


Figure 14: Probability of Exceedance vs Flood Depth for Archetype 14.

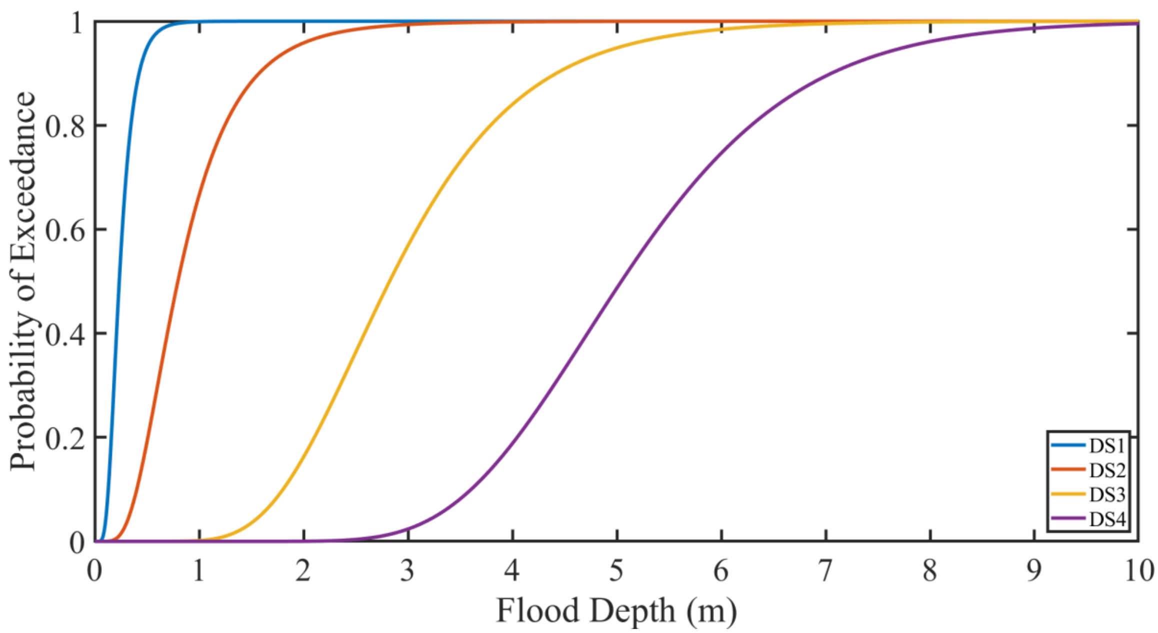


Figure 15: Probability of Exceedance vs Flood Depth for Archetype 15.

Conclusion: This assessment reveals that a significant portion of Nueces County's infrastructure is vulnerable to flooding, especially in areas near the coast. The building damage states, and corresponding probabilities provide valuable insights into the risks faced by individual properties. The results underscore the need for targeted flood management strategies and the implementation of flood protection measures for high-risk areas.

Recommendations:

Regular Risk Assessments: Continuous monitoring and updating of flood risk assessments are necessary to account for changing environmental conditions, population growth, and urbanization.